Daren Chen

Contact Information

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Employment

Caltech

Olga Taussky & John Todd Postdoctoral Scholar Teaching Fellow in Mathematics, Sep 2023 - Present.
Advisor: Yi Ni and Sergei Gukov

Education

Stanford University

Ph.D. in math, Sep 2018 - Jun 2023.
Advisor: Ciprian Manolescu

University of Cambridge, Trinity College

MMath and BA, Sep 2014 - Jun 2018.

Preprints and Papers

1. Applications of the L-space satellite formula(with I. Zemke and H. Zhou), preprint(2025), arXiv:2509.20288
2. The flip map and involutions on Khovanov homology (with H. Yang), preprint(2025), arXiv:2506.00824
3. L-space satellite operators and knot Floer homology (with I. Zemke and H. Zhou), preprint(2024), arXiv:2412.05755
4. A note on rational band moves (with J. Hom, M. H. Kim, J, Park and Z. Wu), La Matematica, Vol.3 (2024), 777-782
5. Bar-Natan Homology for null homologous links in $\mathbb{RP}^3$, Pacific J. Math. 334 (2025), no. 2, 233-268.
6. Adiabatic compactness for holomorphic curves with boundary on nearby Lagrangians (with D. Cant) To appear, Kyoto J. Math, arXiv:2302.13391
7. Floer Lasagna modules from link Floer homology, preprint(2022), arXiv:2203.07650
8. Twistings and the Alexander polynomial, preprint(2022), arXiv:2202.12489
9. Generalised homomorphisms, measuring coalgebras and extended symmetries (with M. Batchelor, W. Boulton, J.Rawlinson, M.Warsi), preprint(2021), arXiv:2105.02527
10. Khovanov-type homologies of null homologous links in $\mathbb{RP}^3$, preprint(2021), arXiv:2104.04779

Talks

1. Two-component L-space links, satellites and the tau-invariant, Topology Seminar, Stanford University, September 2025
2. Two-component L-space links, satellites and the tau-invariant, Geometry/Topology Seminar, UC Davis, May 2025
3. Computing tau-invariant under L-space satellite operations, Topology/Geometry Seminar, University of Oregon, May 2025
4. Flipping map in Khovanov homology, Topology Seminar, UCLA, May 2025
5. Two-component L-space links, satellites and the tau-invariant, Group, Geometry and Topology Seminar, UIUC, April 2025

6. Two-component L-space links, satellite and the tau-invariant Topology Seminar, Georgia Tech, April 2025
7. Computing tau-invariant under L-space satellite operations, Topology Seminar, Michigan State University, March 2025
8. L-space satellite operators and knot Floer homology, JMM Special Session on Concordance and Cobordism in Low Dimensions, Seattle, January 2025
9. L-space satellite operators and knot Floer homology, Geometry and Topology Seminar, UC Irvine, December 2024
10. L-space satellite operators and knot Floer homology, LA Joint Topology Seminar, Caltech, November 2024
11. Formality of Link Floer homology of 2-component L-space links, AMS Sectional Meeting, UC Riverside, October 2024
12. Some variations of Khovanov homology for null homologous links in $\mathbb{RP}^3$, Institute of Mathematics, Chinese Academy of Sciences, May 2024
13. Khovanov homology for null homologous links in $\mathbb{RP}^3$, Caltech-Tsinghua Joint Colloquium, Caltech, March 2024
14. Bar-Natan homology for null-homologous links in $\mathbb{RP}^3$ and genus bound, Geometry/Topology Seminar, Univeresity of South Florida, February 2024
15. Bar-Natan homology for null-homologous links in $\mathbb{RP}^3$ and genus bound, Topology Seminar, Ohio State University, November 2023
16. Some variations of Khovanov homology for null homologous links in $\mathbb{RP}^3$, Virginia Topology Conference, University of Virginia, June 2023
17. Bar-Natan homology for null-homologous links in $\mathbb{RP}^3$ and genus bound, Topology Seminar, Stanford University, April 2023
18. Khovanov-type homologies of null homologous links in $\mathbb{RP}^3$, Geometry/Topology Seminar, Caltech, January 2023
19. Introduction to Floer lasagna module, AMS sectional meeting, University of Utah, October 2022
20. [Online] Khovanov-type homologies of null homologous links in $\mathbb{RP}^3$, Australian Geometric Topology Webinar, September 2022
21. Introduction to Floer lasagna modules, University of Hamburg, August 2022
22. [Online] Twistings and the Alexander polynomial, Knots in Washington 49.75, April 2022
23. [Online] Lasagna algebra and Floer lasagna module, GT GAPS, March 2022
24. Khovanov-type homologies of null homologous links in $\mathbb{RP}^3$, GSTGC 2022, Georgia Tech, March 2022
25. Khovanov-type homologies of null homologous links in $\mathbb{RP}^3$, Topology Seminar, Stanford University, September 2021

Teaching

As a lecturer at Caltech:

- Ma 151a: Algebraic and Differential Topology,
  Autumn 2023, 16 students enrolled, teaching evaluation score: 4.81/5.
  Autumn 2024, 17 students enrolled, teaching evaluation score: 4.53/5.
- Ma 151c: Algebraic and Differential Topology,
  Spring 2024, 13 students enrolled, teaching evaluation score: 4.65/5.
- Ma 157a: Topics in Geometry and Topology,
  Autumn 2024, 3 students enrolled, teaching evaluation score: 5/5.
- Ma 191b: Selected Topics in Mathematics: Morse Theory and Floer Homology,
  Winter 2025, 12 students enrolled, teaching evaluation score: 4.95/5.
- Ma 98: Supervised an undergraduate reading project, Winter 2025
- Supervised a SURF project, Summer 2025

As a teaching assistant at Stanford University:

- Math 20: Calculus, Autumn 2018
- Math 51: Linear Algebra, Multivariable Calculus, and Modern Applications, Winter 2020,Summer 2021, as Admin TA for Winter 2021
- Math 53: Differential Equations with Linear Algebra, Fourier Methods, and Modern Applications, Spring 2023
- Math 116: Complex Analysis, Autumn 2022
- Math 120: Groups and Rings, Autumn 2019
- Math 143: Differential Geometry, Spring 2019
- Math 148: Algebraic Topology, Winter 2023
- Math 215A: Algebraic Topology, Autumn 2020, Autumn 2021
- Supervised a DRP reading project, Spring 2021

Professional Services

Organizer of Caltech Geometry/Topology seminar during the academic year 2024-2025.

Organizing committee for Kylerec 2022, 2023.

Organizer of Student Topology Seminar at Stanford University during the academic year 2021-2022.